

Mandi Mitra

Team Brief — Maharashtra Farm-Gate Price Discovery

Hackathon	Maharashtra State Innovation Society
Problem ID	26132 — Strengthening market linkages and price discovery for farmers
Department	Maharashtra State Innovation Society, Dept. of Skills, Employment, Entrepreneurship a
Theme	Agriculture, FoodTech & Rural Development
Category	Software
Deadline	Internal hackathon in 4 days
Goal	70% functional prototype, 100% spec acknowledged

Our Team Situation

We're vibe-coding this in 4 days for the internal hackathon. We picked problem statement 26132 (Maharashtra farm-gate price discovery) and scoped it down hard. The full PS asks for 16 features covering market intelligence, logistics, storage, payments, and dispute resolution. We're not naive about that — we've explicitly triaged what we build vs. what we acknowledge on a Roadmap page.

Our constraint: a 70% functional prototype, not a full product. That means the farmer-side decision-support flow (prices + chart + sell/wait rec) and the buyer-side lot + offer flow work end-to-end. Logistics, escrow, KYC, and dispute processes show up as 'Coming Soon' on a dedicated page — which signals to judges that we read the full spec, even where we deferred.

Stack is plain Next.js + shadcn + Recharts with JSON-file storage. No database, no auth, no payment gateway. Marathi + English toggle is included because Maharashtra's farmers are the actual user, and a Marathi UI is a sharp differentiator at the internal review.

If anything below is unclear or you disagree with the scope, raise it before Day 1 setup — easier to redirect now than mid-build.

What We're Building

A web app called **Mandi Mitra** (Friend of the Market) that helps small farmers in Maharashtra make better selling decisions. Right now, a farmer has 3 buyers in front of him and **zero** way to know which is offering a fair price. He just picks the easiest option and loses money. We fix that.

The Big Idea

Don't build a full marketplace. Build a decision-support tool.

The one question a farmer needs answered: *Should I sell today or wait?*

Our app shows him:

- Today's price at 3 nearby mandis (side by side)
- 30-day price trend chart
- A clear recommendation: **SELL NOW / WAIT 3 DAYS / WAIT 2 WEEKS**

- 2–3 verified buyers who want his crop right now

Plus a basic two-sided flow: farmer posts a lot, buyer makes an offer, deal is recorded.

Who Uses It

PRIYA — Farmer	Opens app on phone Selects: Soybean, Latur district Sees: prices, chart, recommendation, buyers Creates a lot: 10 tons, Grade A, asking 4,400/quintal
RAJAN — Buyer	Opens app on desktop Browses lots, sees Priya's soybean lot Makes offer: 4,350/quintal for 10 tons Priya accepts — transaction shows up in history

Features We're Delivering

The problem statement asks for 16 specific features. Here's our plan for each:

BUILT	Fully functional feature, end-to-end working
MOCKED	Visible in UI, data is seeded (looks real)
DEFERRED	Acknowledged on a 'Roadmap' page, not built

#	Feature	Status
A	Mandi price aggregation	BUILT
B	Buyer demand aggregation	BUILT
C	Quality requirements per lot/buyer	BUILT
D	Arrival volumes at mandi	MOCKED
E	Transport options	DEFERRED
F	Storage options	DEFERRED
G	Localised price trends	BUILT
H	Sale-window recommendations	BUILT
I	Match farmers/FPOs with buyers	BUILT
J	Verified buyer credentials	MOCKED
K	Lot creation	BUILT
L	Quality grading (A/B/C)	MOCKED
M	Digital offers	BUILT

N	Logistics coordination	DEFERRED
O	Payment tracking (escrow)	DEFERRED
P	Dispute / grievance process	DEFERRED

Tally: 8 BUILT · 3 MOCKED · 5 DEFERRED | Functional: 8/16 (50%) | Acknowledged: 16/16 (100%)

Tech Stack

Framework	Next.js (App Router) + TypeScript
Styling	Tailwind CSS + shadcn/ui components
Charts	Recharts
Languages	next-intl (English + Marathi toggle)
Storage	JSON files in /data (simple, fast, no DB setup)
Approach	Vibe-coding — pick tools we're comfortable with

Data Plan

Crops	Soybean, Onion, Tur
Mandis	Latur, Pune, Nashik, Solapur, Nagpur
Day 1 priority	Create a realistic seed dataset (60 days of prices). Even if we successfully scrape Agmarknet later

4-Day Plan

Day 1	Setup, data, basic structure, seed dataset
Day 2	Farmer dashboard — HERO feature : price + chart + recommendation
Day 3	Buyer dashboard + lot + offer flow + transactions
Day 4	Polish + demo prep + rehearsal + Loom backup

See **CHECKLIST.md** for the full task list.

Roles (suggested)

Person 1 Architecture	Data layer, Agmarknet scraper, recommendation engine, API routes
Person 2 Farmer UI	Farmer dashboard, charts, Marathi translations, mobile responsive
Person 3 Buyer UI	Buyer dashboard, lot + offer flow, transactions page, polish

Day 1: all three sit together and align on the setup. After that, work in parallel and review each other's code.

Demo Script (3 minutes)

0:00 – 0:30	Problem: farmer has no price info, loses 15%
0:30 – 1:30	Farmer flow: open app, see prices + chart + recommendation
1:30 – 2:30	Marketplace: create lot, buyer makes offer, accept
2:30 – 3:00	What's next: roadmap, Marathi support, outcomes

The whole thing ends with a transaction record visible. That's our *proof it works* moment.

Judge-Facing Outcomes

The problem statement asks for these outcomes. Here's how we show each one:

- **Improved farmer price realisation** — Sell/wait rec + buyer competition visible
- **Reduced information asymmetry** — One screen: 3 mandis + 3 buyers + recommendation
- **Lower transaction cost** — No-middleman direct farmer-to-buyer lot flow
- **Stronger FPO aggregation** — FPO persona in the data
- **Reduced post-harvest loss** — Sell/wait timing advice reduces distress sales
- **More reliable buyer sourcing** — Verified badge + transaction history
- **Transparent transaction records** — /transactions page

Ground Rules

- Don't add features past Day 3. Polish beats more features.
- Don't skip the seed data. Always have a working demo.
- Marathi on key screens, not error messages. Don't over-translate.
- Mobile-responsive. Judges will check.
- Test the demo URL with ?crop=soybean&district=latur pre-loaded.
- Have a 3-min Loom backup recorded on Day 4, in case live demo dies.